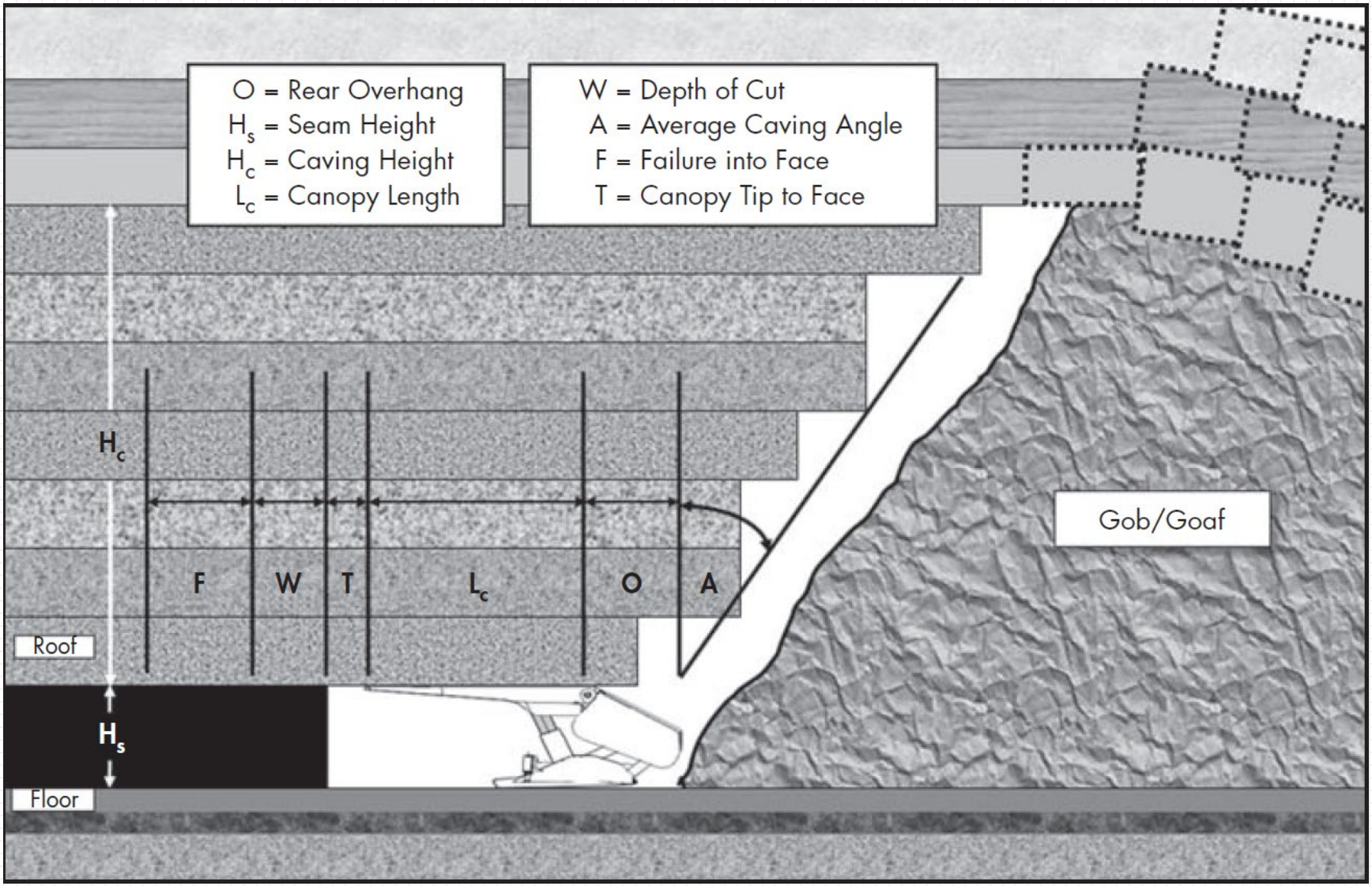
NOVEMBER	TOPIC
11	U/G MINING - EXTRACTION

Strata gases

- Methane (CH_4)
- Carbon Monoxide (CO)
- Carbon Dioxide (CO_2)
- Hydrogen Sulphide (H_2S)

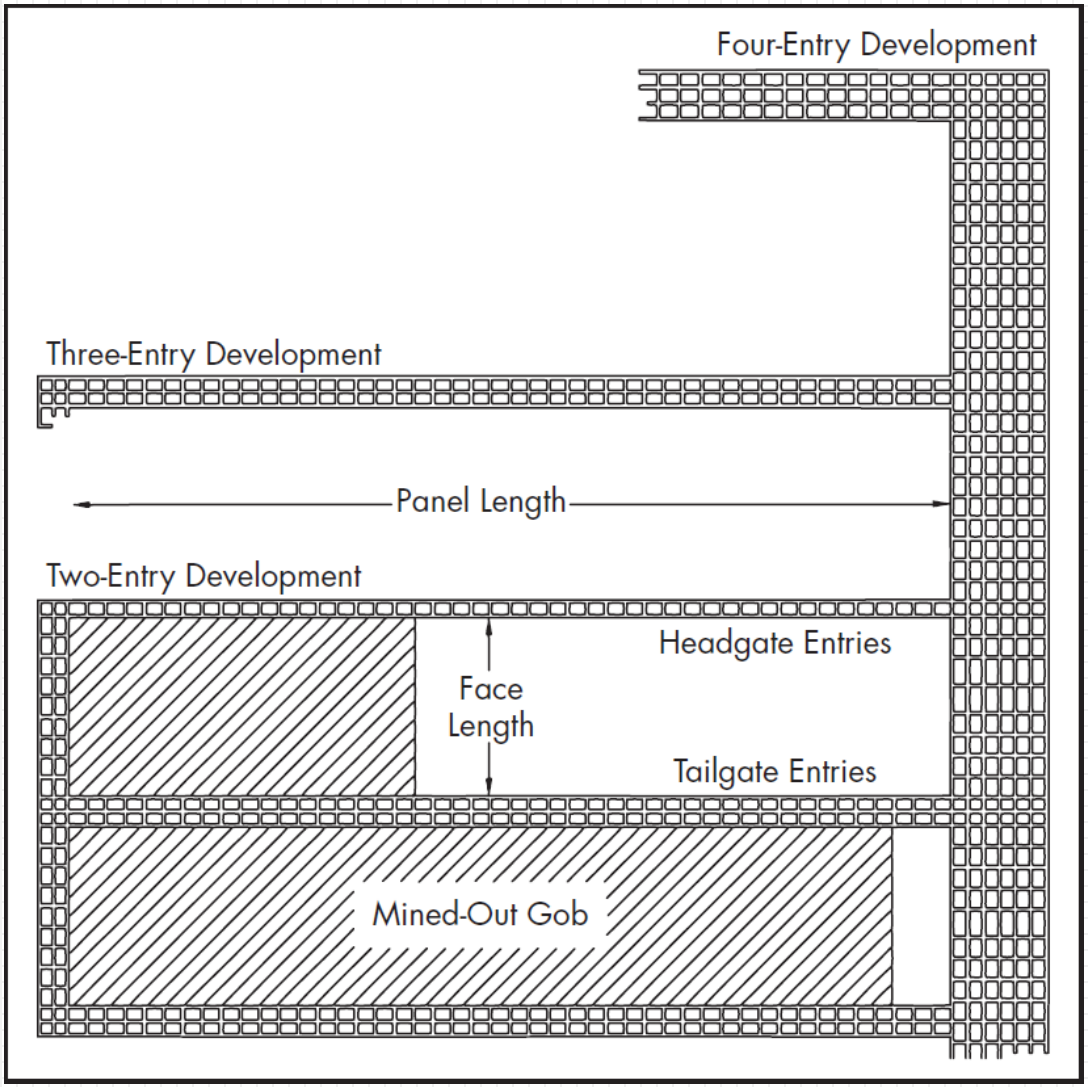
Conceptual loading model for longwall



Factors affecting roof loading

- Seam thickness (extraction height),
- Near-seam strata (type and thickness),
- Seam and nearby strata material properties,
- Caving behavior of strata,
- Seam inclination,
- Natural and mining-induced fractures, and
- Previous mining above or below the seam.

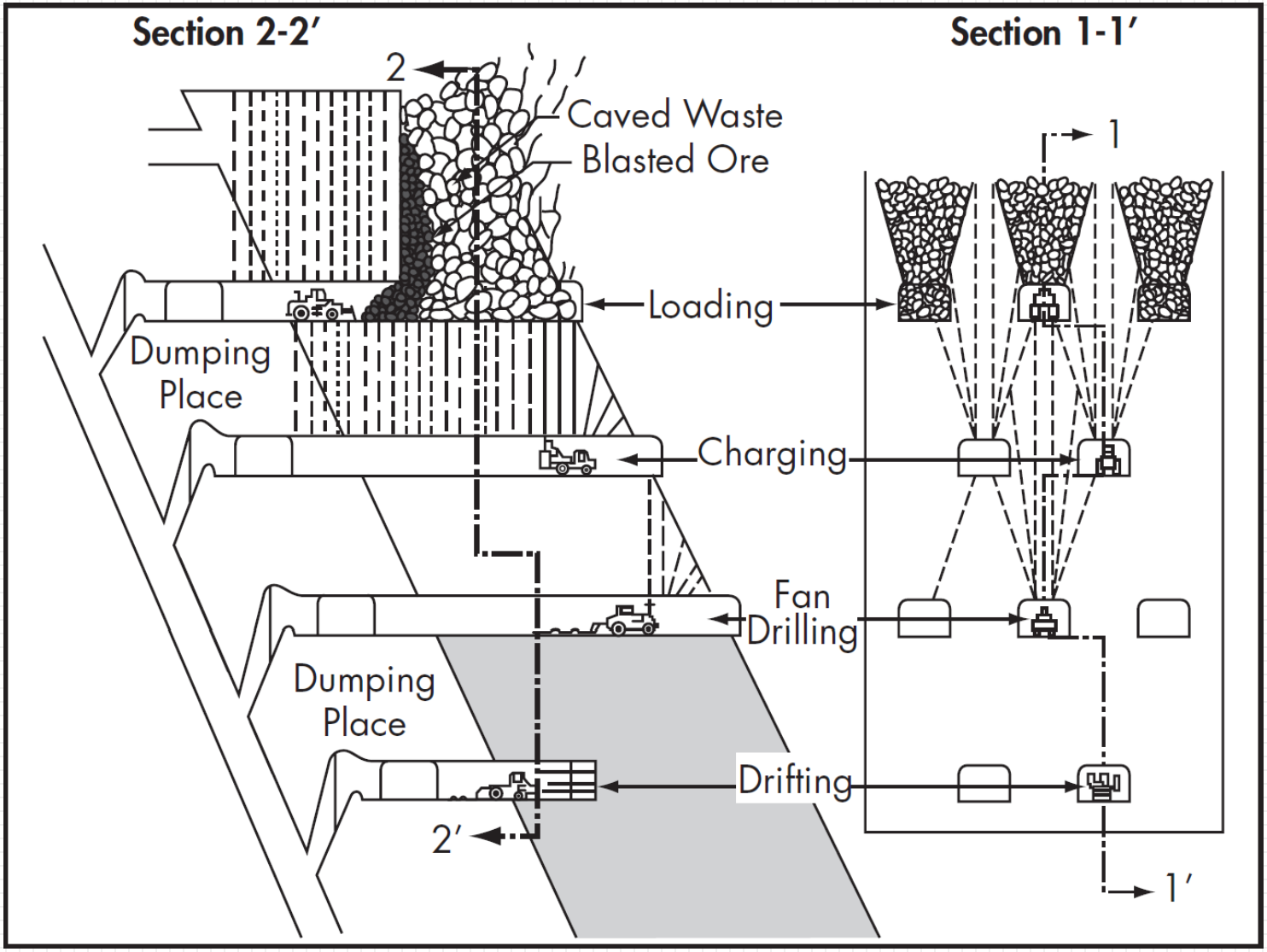
Fully retreating longwall layout



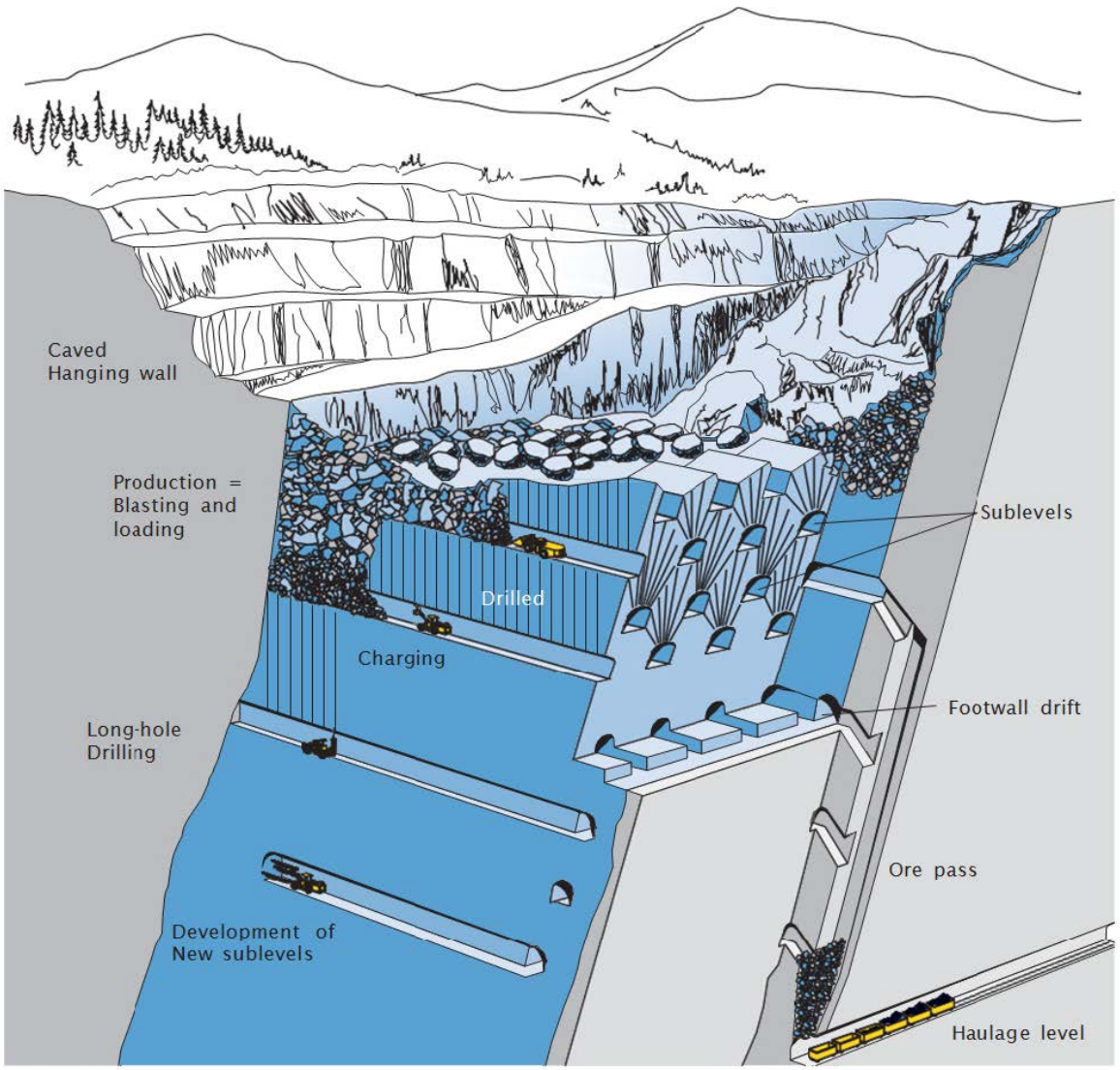
NOVEMBER	TOPIC
11	U/G MINING - EXTRACTION

SUBLEVEL CAVING

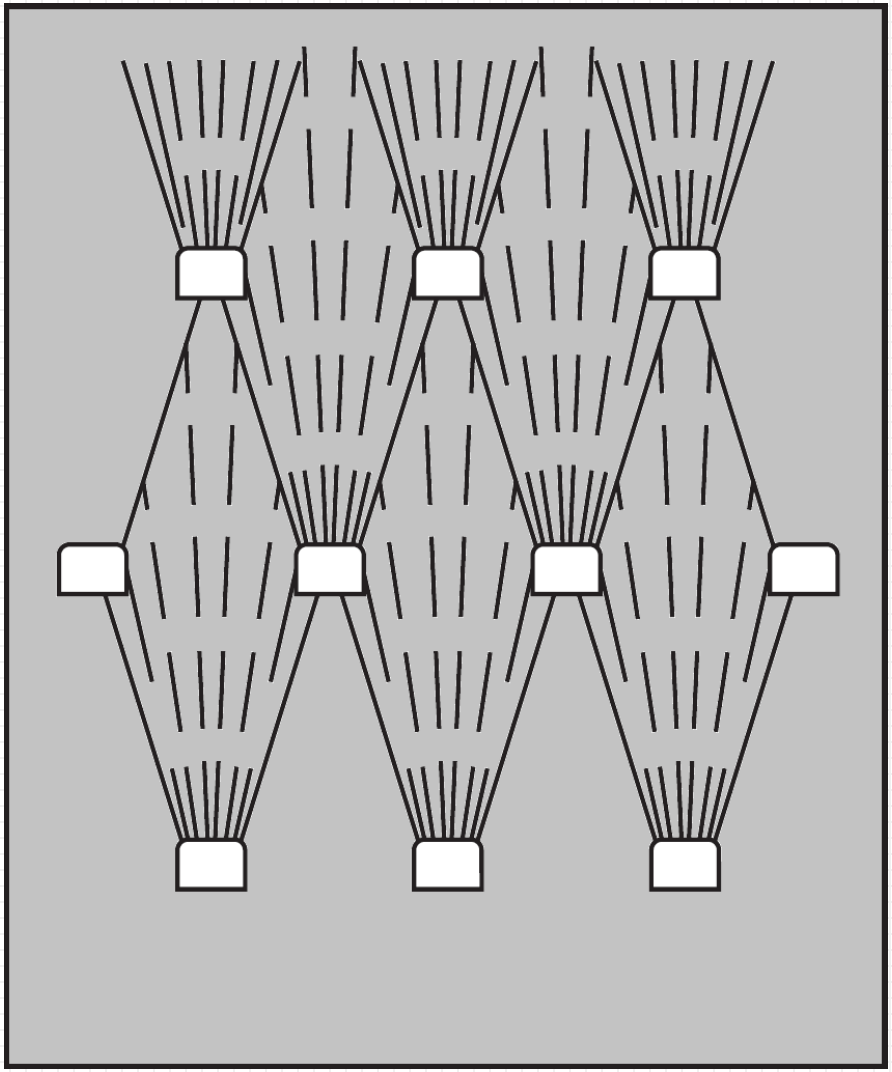
Schematic of sublevel caving mining

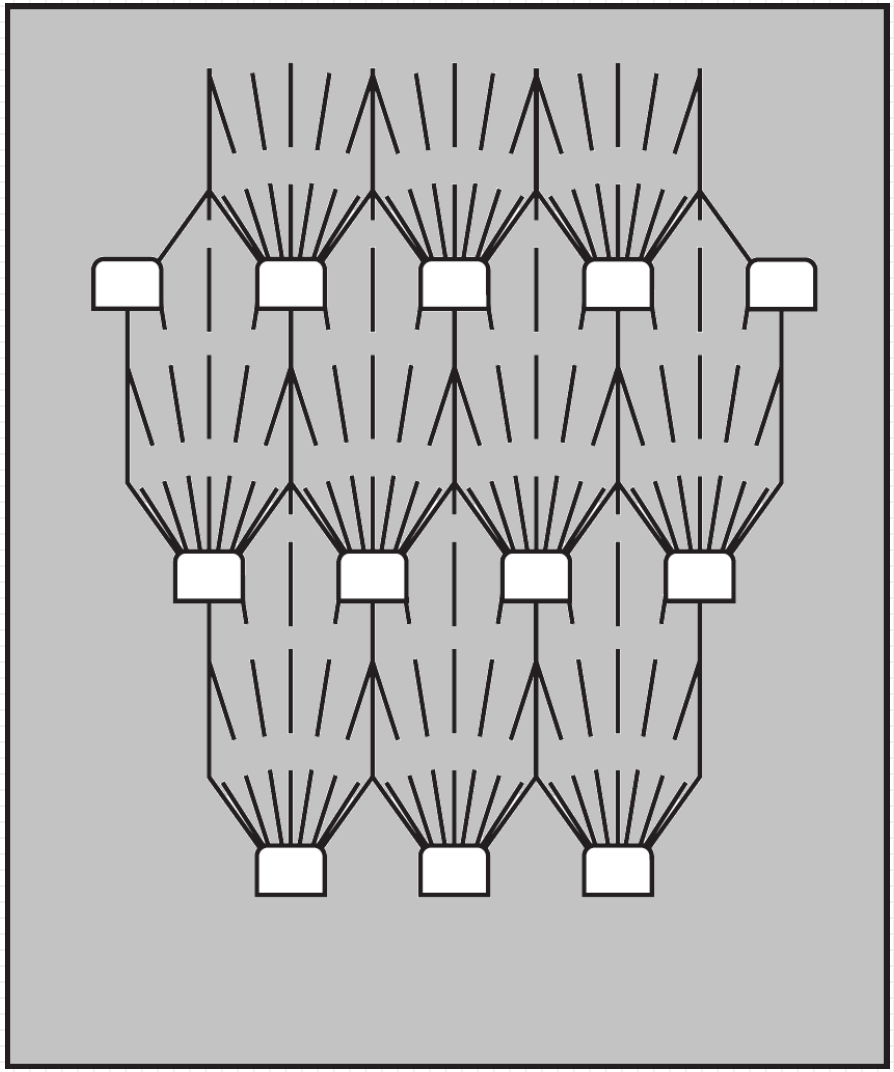


Schematic of sublevel caving mining



Sectional view of sublevel caving ring options



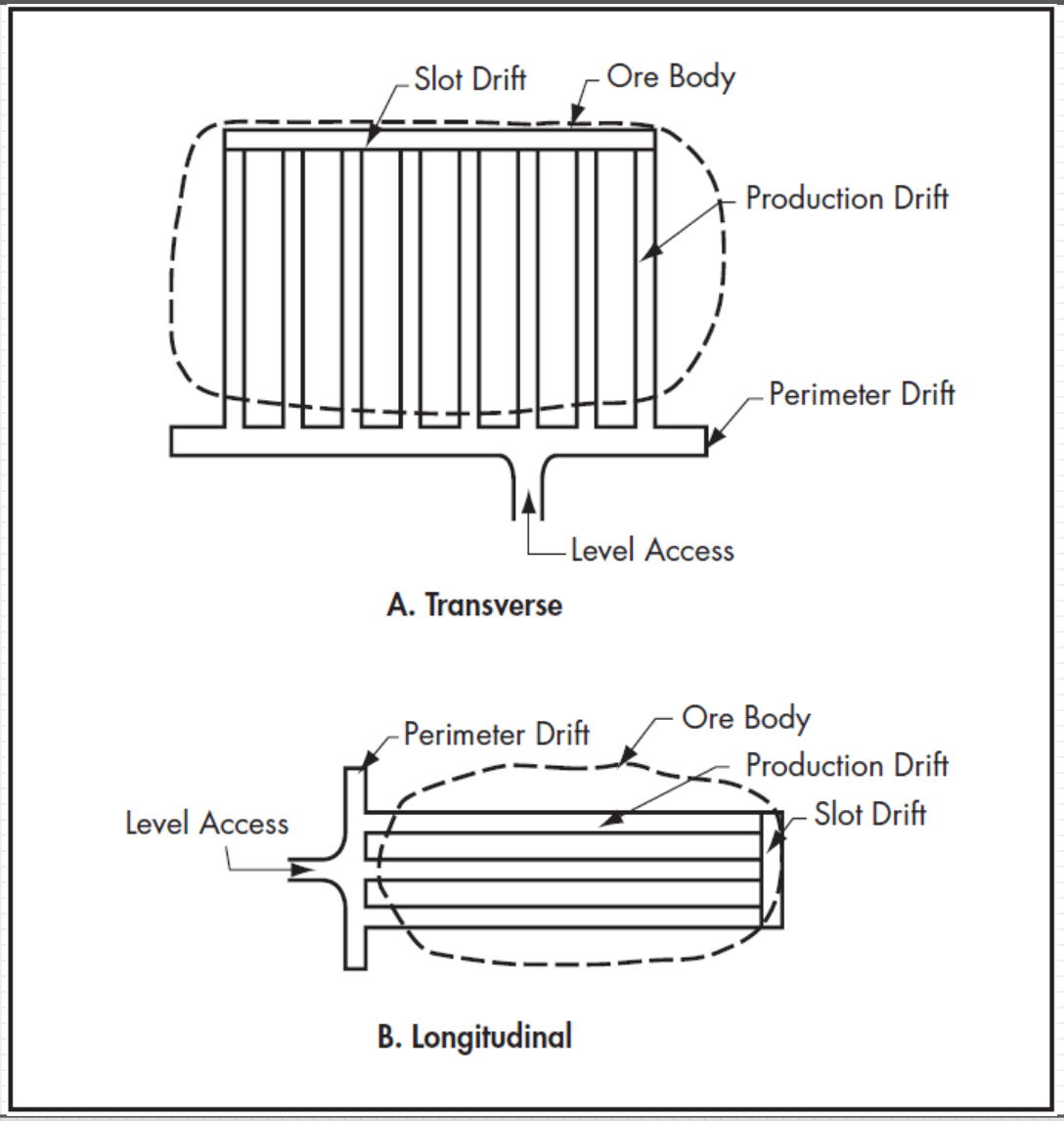


The suitable ore-body characteristics

- Cavable overlying host rock,
- Relatively large ore-body or a relatively weak hanging wall,
- Steeply dipping and relatively uniform ore body,
- Visually different ore and waste properties.

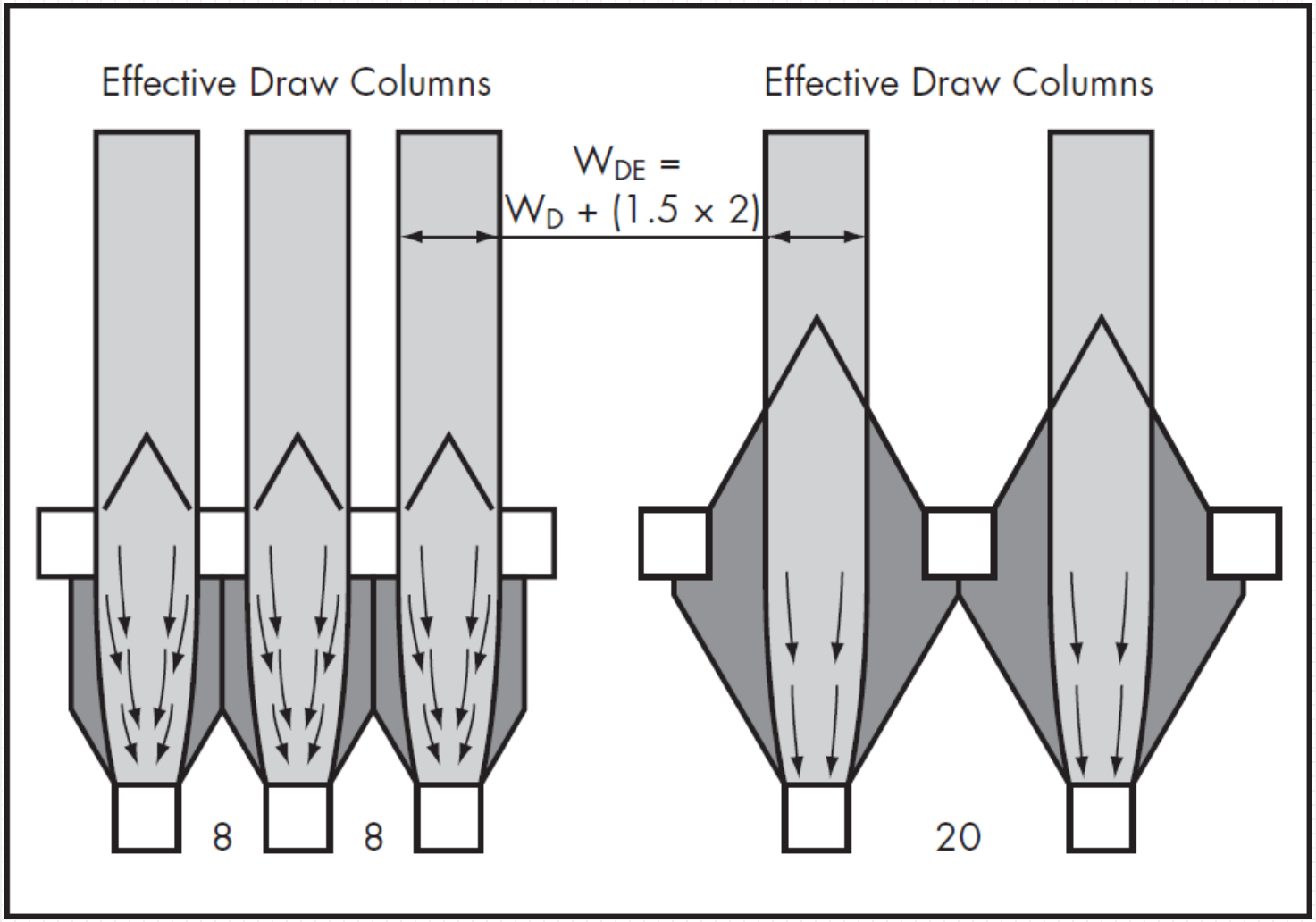
Parameters producing significant variability in sublevel caving systems

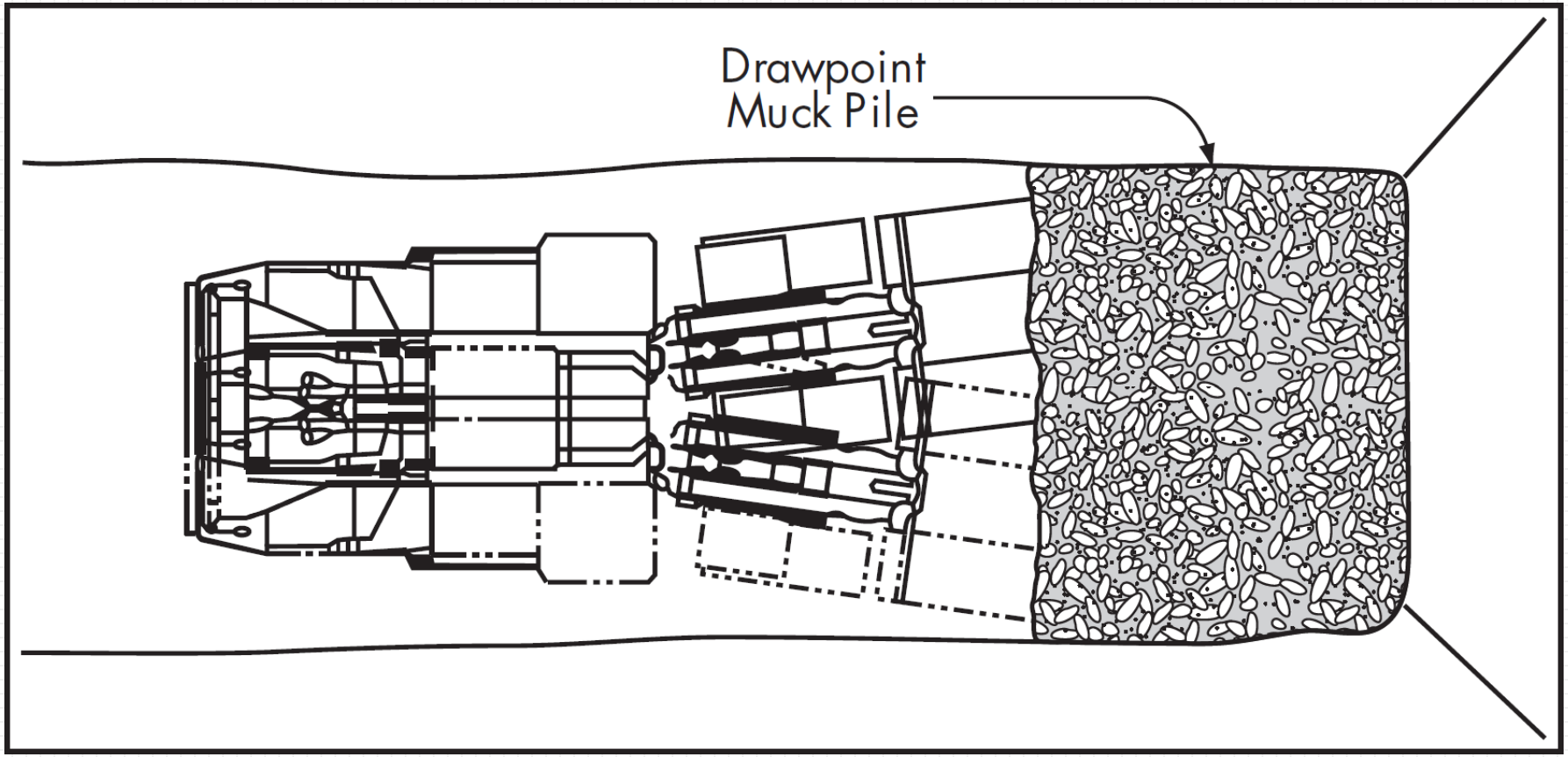
- Rock mass properties,
- Blast design,
- Ability of charging crew to implement blast design (blocked holes, etc.),
- Hole deviation,
- Blasting performance of rings around the current ring,
- Hang-up occurrence impacting draw, and
- Grade variations

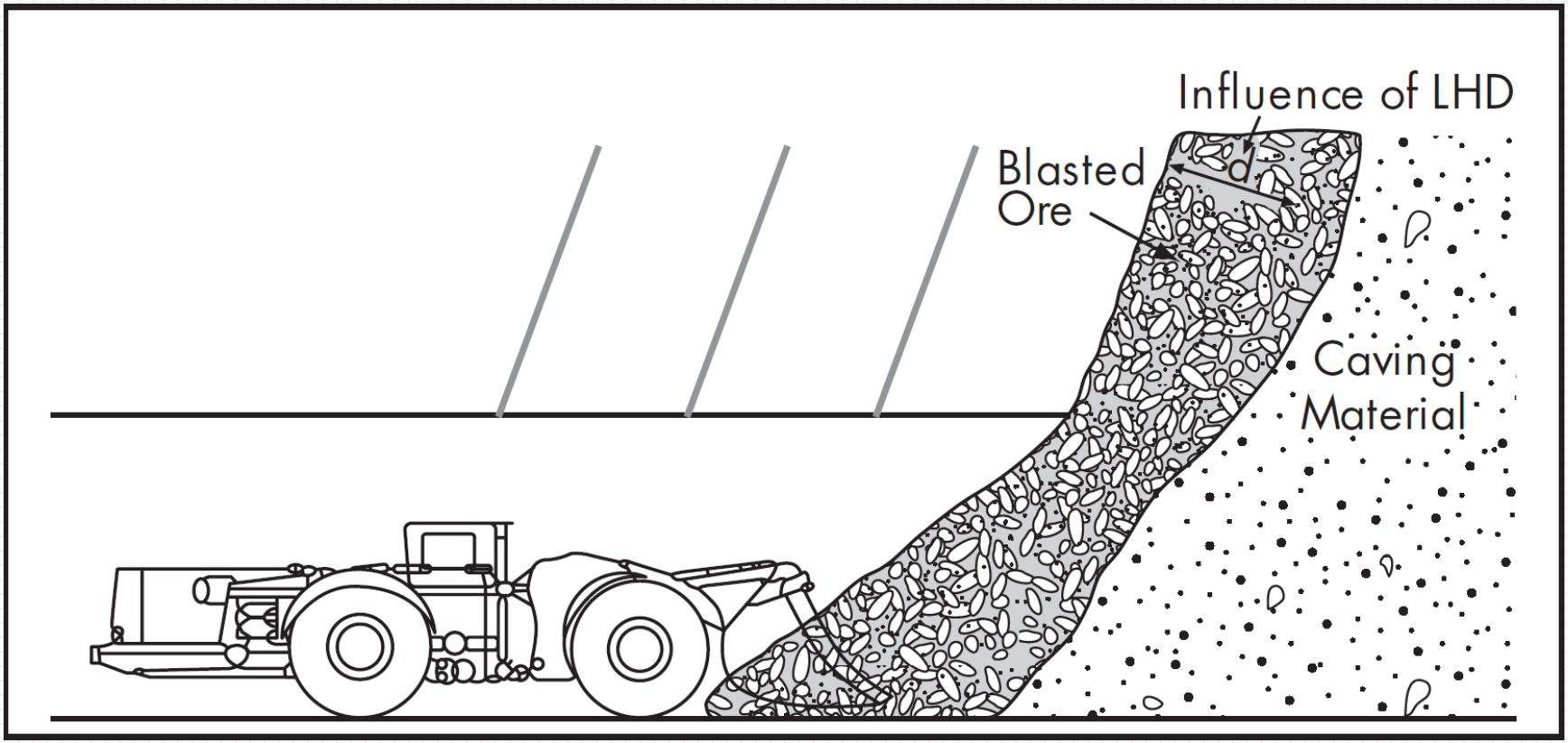

Slot Drift
Ore Body
Production Drift
Perimeter Drift
Level Access
A. Transverse
Perimeter Drift
Ore Body
Production Drift
Slot Drift
Level Access
B. Longitudinal

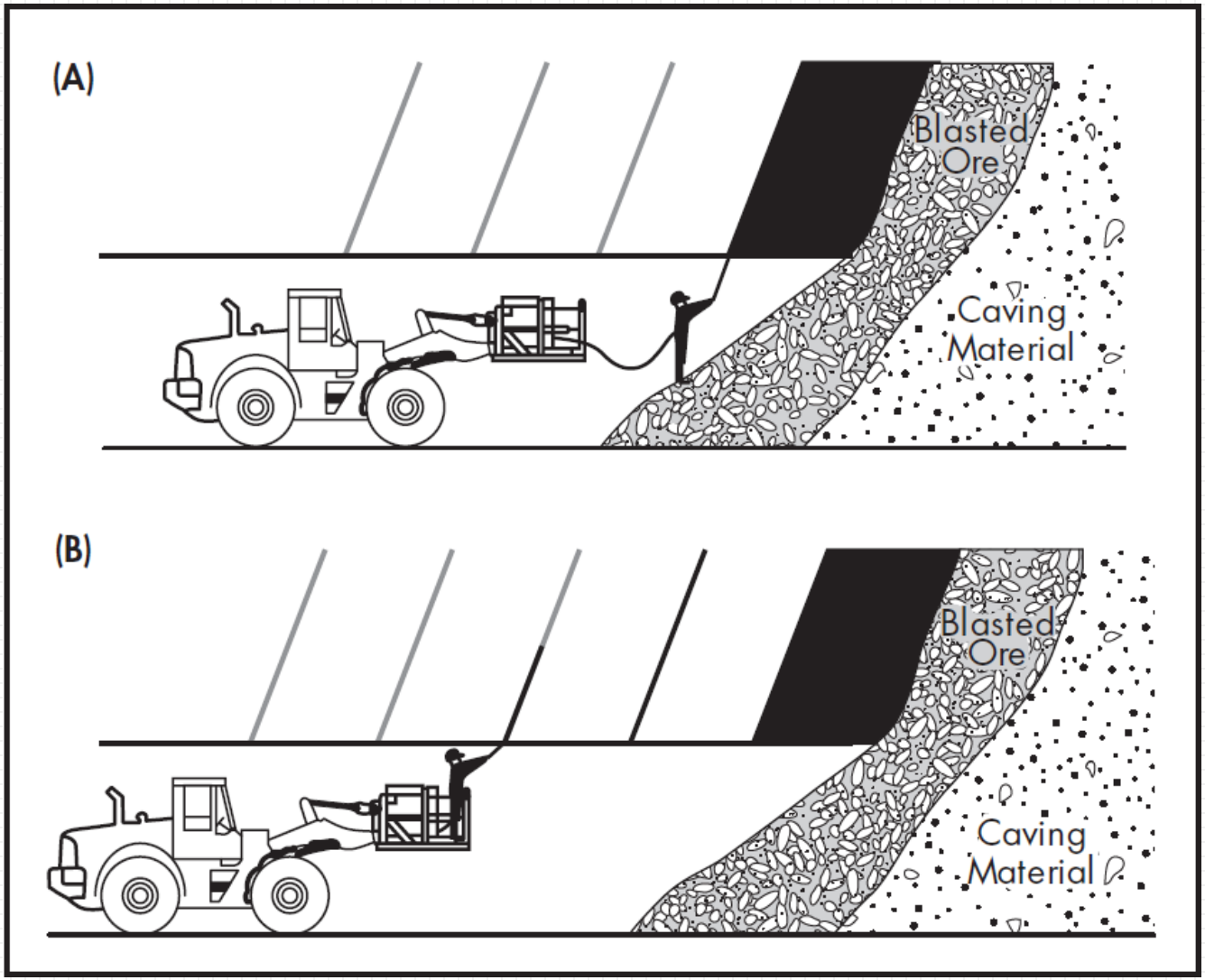
Important factors in selection of production drift spacing

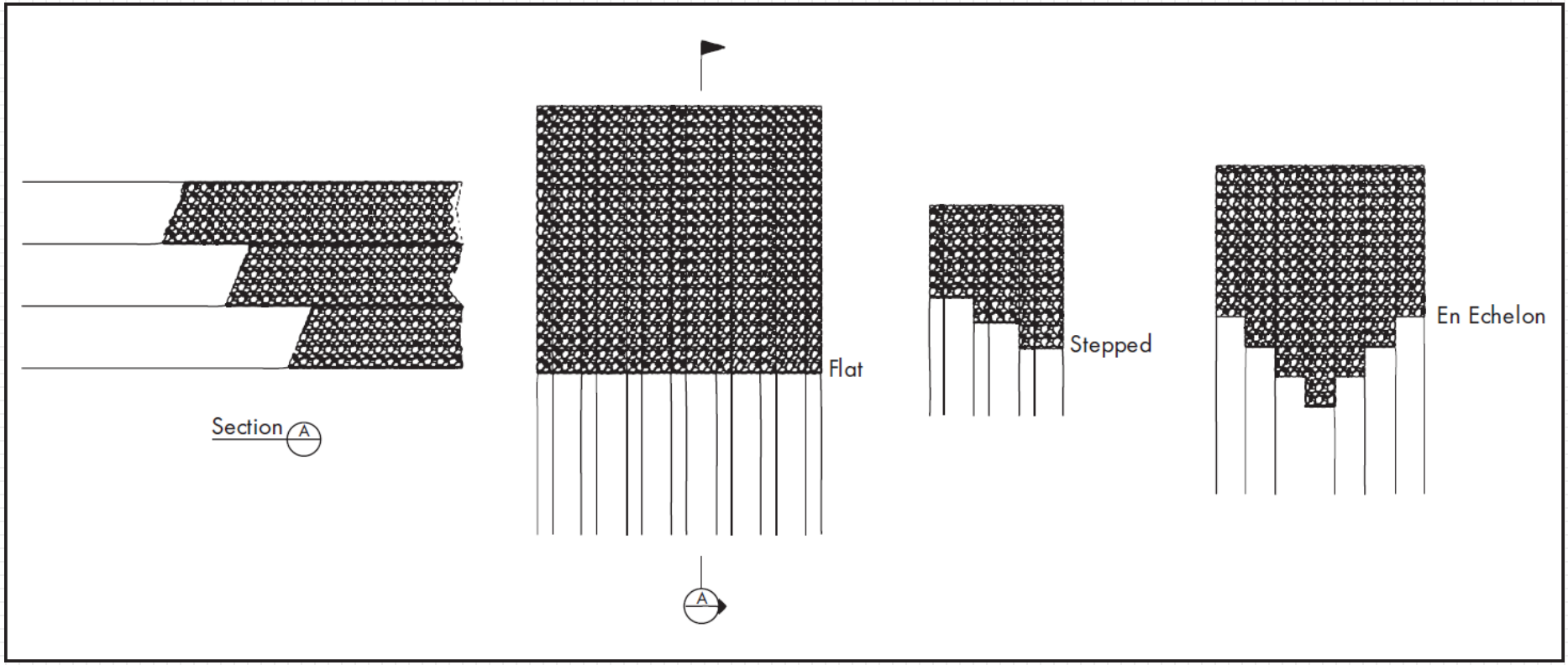
- Potential ring shape produced by the combination of sublevel height and production drift spacing
- Granular flow properties inherent to the mine
- Value of the mineral being mined
- Geotechnical factors
- Production drilling and charging factors.

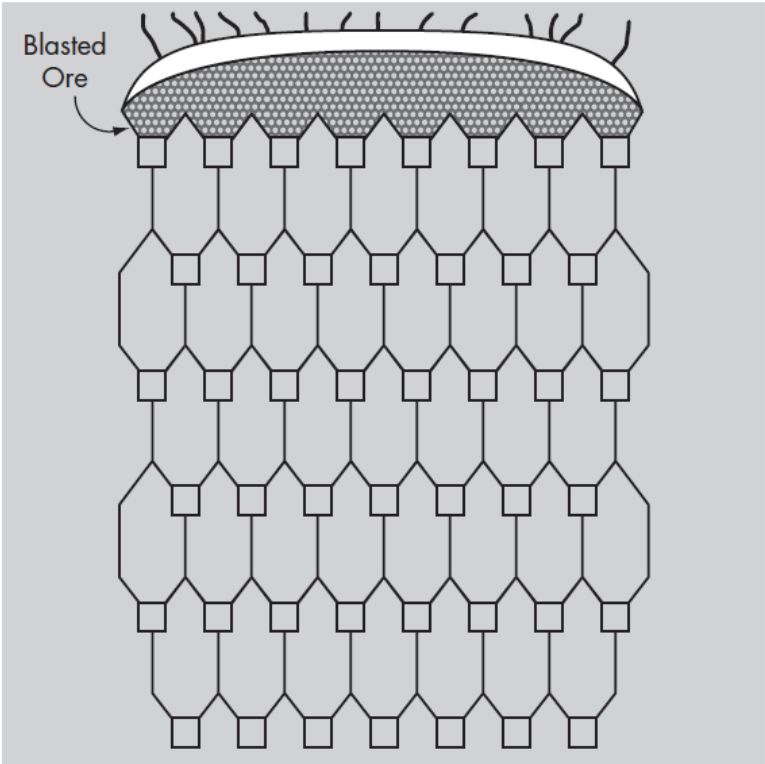

Effective Draw Columns
Effective Draw Columns
$W_{DE} = W_D + (1.5 \times 2)$
8
8
20


Drawpoint
Muck Pile

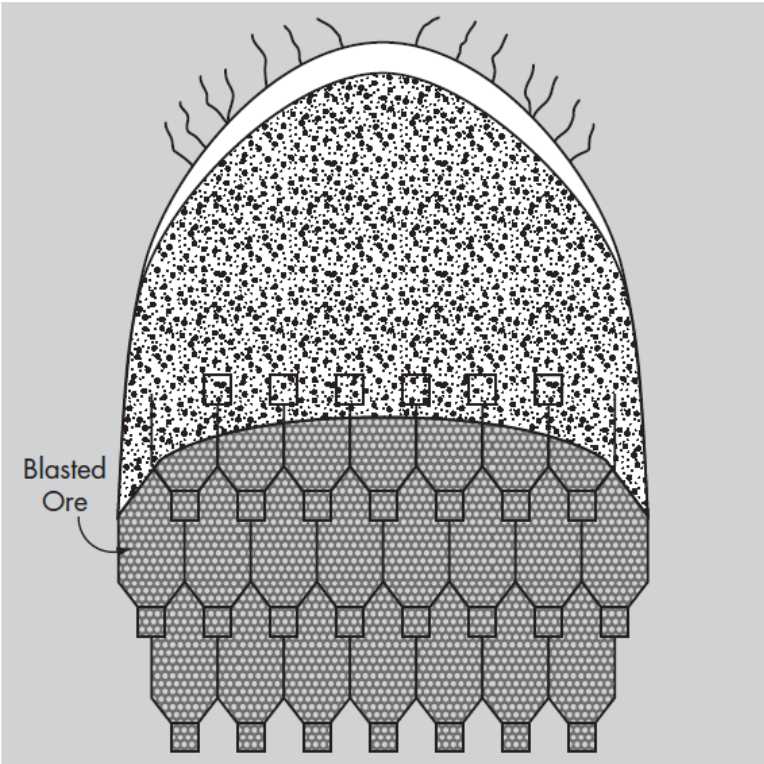

Influence of LHD
d
Blasted
Ore
Caving
Material


(A)
Blasted Ore
Caving Material
(B)
Blasted Ore
Caving Material

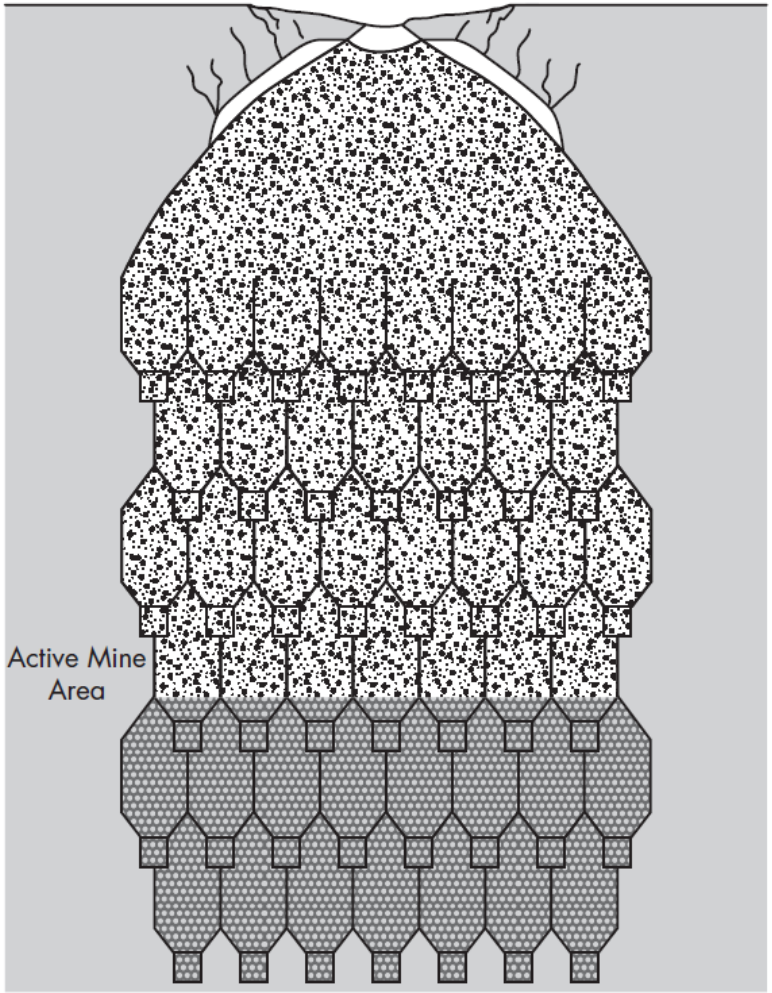

Section A
Flat
Stepped
En Echelon
A



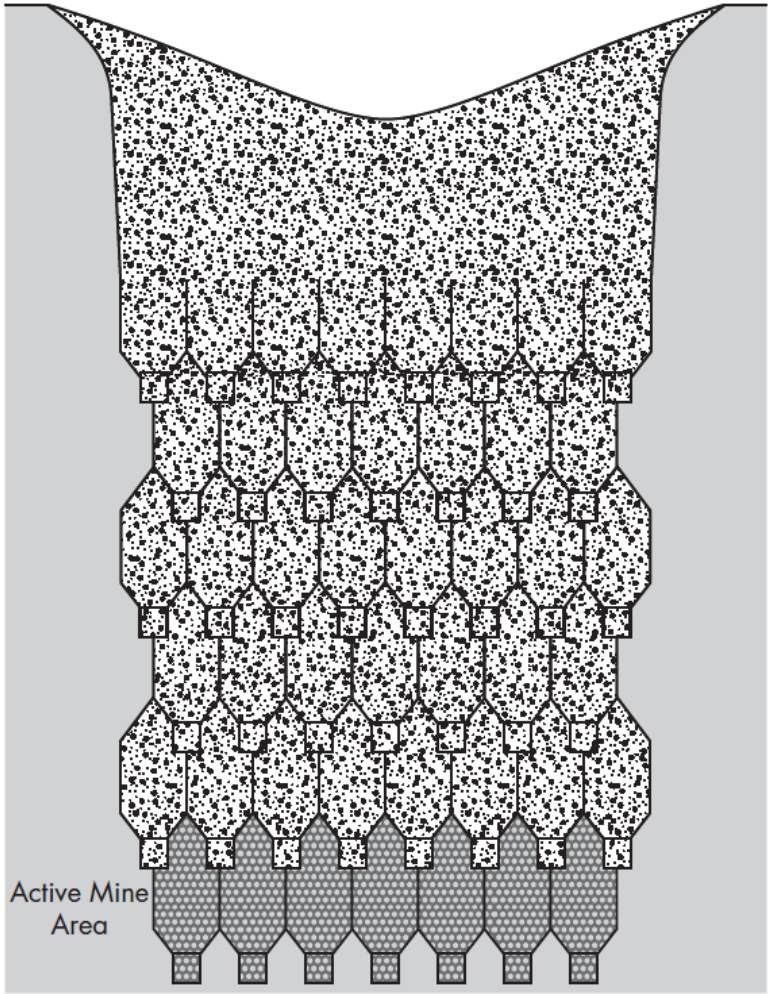
A. Initial undercut level



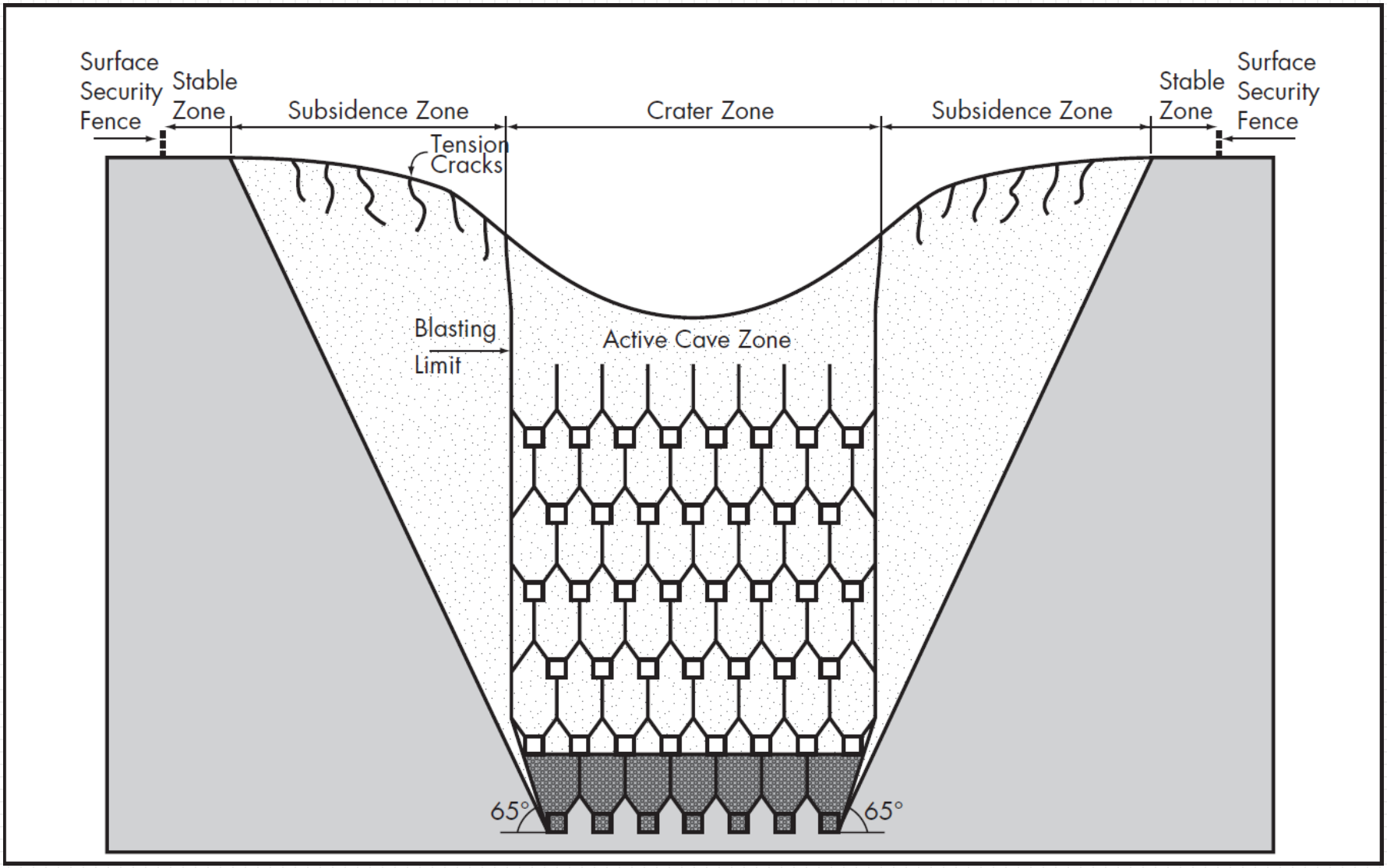
B. Cave propagation toward surface



C. Initial surface subsidence as the cave breaches the surface



D. Surface crater created as the sublevel caving operation deepens


Surface Security Fence
Stable Zone
Subsidence Zone
Crater Zone
Subsidence Zone
Stable Zone
Surface Security Fence
Tension Cracks
Blasting Limit
Active Cave Zone
65°
65°